

What the Product Formula Allows: A Rigidity Theorem and an Effective-Weight Test for the ATS Height Subchain

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2 October 2026 — Version 0.3 (draft)

Abstract

Part 5 of this series [15] reports a source-bound ATS substance audit; this corrective Part 6 states the exact scope of its effective-weight test. For a fixed number field L , its fixed standard absolute values, and the full product formula tested on every $x \in L^\times$, a family of real exponents (γ_v) must be constant across all places, independently of the class number. Finite-fiber redistribution gives the same conclusion for effective weights. A prescribed j^2 -weight on a nonempty place set admits the constant global extension; only unequal final place weights are excluded. For global quantity families with a proved linear effective-weight representation, the product formula leaves at most a y -dependent global scalar. Mere dependence on the weights is insufficient for class membership. Under ATS II($\frac{1}{2}$) (5.3.3), direct substitution identifies the weighted degree on $L^\times$ and the projective height on $\mathbb{P}^n(L)$ with their classical values; the source's stabilization on $L^\times$ is then constant term by term. The extension to $\mathbb{P}^n(\overline{L})$, the ATS III line-bundle height, and the ATS IV volume apparatus require separate bridges. Log-volume and measure membership or nonmembership in the linear class is unclassified here. Coordinate hyperplanes and their pullbacks are distinct objects. No global closure of the ATS repair program, proof or refutation of abc/IUT, or mathematical certification of the whole ATS series is claimed.

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1 Mandate, Scope, and Source Versions

This paper is Part 6 of the abc Gap Series. Part 1 [9] localized the quantitative gap in the transport step of Corollary 3.12 of IUTchIII [8]; Part 2 [14] supplied a form criterion; Part 3 [10] distilled a checkable admissibility standard; Part 4 [11] audited the ATS series [3, 4, 5, 6, 7] at ledger level (v1-F). The present paper reports the mathematical core of the repair program: a rigidity theorem, a factorization instrument, and their source-bound application to the *height subchain* of the ATS texts.

Scope discipline. “The audited passages” means: ATS II($\frac{1}{2}$) §§5.3, 5.10, 8.2–8.7 and ATS IV §1.5, in the exact arXiv versions cited in the bibliography (II($\frac{1}{2}$): v12; IV: v2; both revised 24 February 2025); ATS III v4, Remark 8.9.2(2), is a separately typed application target. These are publicly posted, versioned preprints, not peer-reviewed publications; ATS IV carries the header “Preliminary version for comments.” This paper does not audit the full ATS series, does not classify its volume and measure apparatus (see Case C of Section 6), and makes no claim about structures outside the audited passages. Ground rules as in Parts 4–5: no authority argument, no truth verdict, verbatim anchors; findings about later program work appear only as dated external status notices (Section 7), separate from the proofs of this paper.

2 Types and Operational Glossary

Fix a number field L and its full set V_L of archimedean and finite places. The local inputs and global quantity families have different types:

- b_v is the fixed standard absolute value: $|\sigma(x)|$ at a real place, $|\sigma(x)|^2$ at a complex place, and $N\mathfrak{p}^{-v_{\mathfrak{p}}(x)}$ at a finite place. Thus $\prod_v b_v(x) = 1$ for every $x \in L^\times$.
- $\rho_{y,v} = b_v^{a_{y,v}}$, with $a_{y,v} > 0$, is the equivalent source absolute value on L_v . We abbreviate the source datum by y .
- $\lambda_{y,v}$ is a normalization exponent. In our notation, ATS II($\frac{1}{2}$) (5.3.3) [5] says $b_v = \rho_{y,v}^{\lambda_{y,v}}$. This is a paraphrase with renamed symbols, not a verbatim quotation of the source’s α/ξ notation.
- $\beta_{y,v} = a_{y,v}\lambda_{y,v}$ is the effective weight. The pointwise identity (5.3.3) implies $\beta_{y,v} = 1$ for every v ; it also implies $\lambda_{y,v} > 0$.

The rigidity theorem constrains effective weights, not the two factors separately. Coordinate variation therefore need not change metric values.

For a precise coordinate comparison use the common finite-support space $\mathbb{R}^{(V_L)}$ and

$$H_y = \{t \in \mathbb{R}^{(V_L)} : \sum_v \lambda_{y,v} t_v = 0\}, \quad \ell_y(x) = (\log \rho_{y,v}(x))_v.$$

The vector $\ell_y(x)$ has finite support for $x \in L^\times$. The normalization identity gives $\ell_y^{-1}(H_y) = L^\times$. Nonproportional coefficient families define different kernels in this common space, while these pullbacks coincide. This conditional linear algebra observation neither proves nonproportionality for a particular source parameter pair nor verifies the entirety of Theorem 5.10.1.

A family *factors through effective weights* here only when it has the linear representation in Definition 4.1. A *classical* value is one obtained from the fixed standard family on the stated arithmetic domain. A *place-selective* weight family has unequal final weights, rather than a single global scalar.

Table 1: Local inputs, coordinates, and global quantity families.

Object	Type and dependence	Relation to $\mathcal{G}_{\text{eff}}$
b_v	Fixed local reference	Input, not a global family
$\rho_{y,v}$	Possibly varying local absolute value	Input
$\lambda_{y,v}$	Normalization coordinate	Parameter
$\beta_{y,v}$	Effective coefficient; identically 1 under (5.3.3)	Coefficient
H_y	Coordinate kernel; variation conditional on coefficients	Coordinate object
$h_y, \deg_y$	Global values on $\mathbb{P}^n(L)$ and $L^\times$; invariant under (5.3.3)	Verified linear representations
vol_y, μ_y	Concrete measure objects and dependence require identification	Membership and nonmembership unclassified

3 The Rigidity Theorem

Theorem 3.1 (Rigidity of product-formula weights). *Let L be a number field and $(\gamma_v)_{v \in V_L}$ a family of real exponents such that*

$$\prod_{v \in V_L} b_v(x)^{\gamma_v} = 1 \quad \text{for all } x \in L^\times.$$

Then γ_v is constant in v . (We write γ for the generic exponent family of the theorem, reserving λ for the source's normalization exponents of Section 2.)

Proof. Set $\mu_v := \gamma_v - 1$. Subtracting the logarithm of the standard product formula from the logarithm of the hypothesis gives

$$\sum_{v \in V_L} \mu_v \log b_v(x) = 0 \quad \text{for all } x \in L^\times,$$

where for each fixed x all but finitely many terms vanish.

(i) *Archimedean component.* For units $u \in \mathcal{O}_L^\times$ all finite terms vanish, so $\sum_{\sigma \text{ arch}} \mu_\sigma \log b_\sigma(u) = 0$ for all u . By Dirichlet's unit theorem [2, Ch. I, §7], $\text{Log}(\mathcal{O}_L^\times)$ is a full lattice in the trace-zero hyperplane $H = \{y \in \mathbb{R}^{r_1+r_2} : \sum_\sigma y_\sigma = 0\}$: it lies in H because $\sum_\sigma \log b_\sigma(u) = \log |N(u)| = 0$, and its rank is $r_1 + r_2 - 1 = \dim H$. The vector $(\mu_\sigma)_\sigma$ annihilates a full lattice of H , hence all of H , and therefore lies in $H^\perp = \mathbb{R} \cdot (1, \dots, 1)$:

$$\mu_\sigma = c \quad \text{for all archimedean } \sigma, \text{ for some } c \in \mathbb{R}.$$

(ii) *Finite component.* Let $\mathfrak{p}$ be any prime ideal and h the order of its class in the finite class group [2, Ch. I, §6], so $\mathfrak{p}^h = (x_\mathfrak{p})$ is principal. Then $\log b_\mathfrak{q}(x_\mathfrak{p}) = 0$ for all finite $\mathfrak{q} \neq \mathfrak{p}$, while $\log b_\mathfrak{p}(x_\mathfrak{p}) = -h \log N\mathfrak{p}$ and

$$\sum_\sigma \log b_\sigma(x_\mathfrak{p}) = \log |N(x_\mathfrak{p})| = \log(N\mathfrak{p}^h) = h \log N\mathfrak{p}.$$

Substituting into the annihilation identity and using (i):

$$\mu_\mathfrak{p} \cdot (-h \log N\mathfrak{p}) + c \cdot (h \log N\mathfrak{p}) = 0 \quad \implies \quad \mu_\mathfrak{p} = c.$$

As $\mathfrak{p}$ was arbitrary, $\mu_v = c$ for all v , i.e., $\gamma_v = 1 + c$ is constant. Conversely, every constant family satisfies the hypothesis. $\square$

Remark 3.2 (Status). The theorem is elementary and lies in the orbit of the Artin–Whaples axiomatics [1]; we state and prove it in full because the series uses it as a typed test instrument, and because the consequence drawn in Sections 4–6 depends on its exact hypotheses. No novelty claim is made for the theorem itself.

Corollary 3.3 (Unequal final place weights are excluded). *Under Theorem 3.1's same-field, full-place hypothesis, only constant final exponent families satisfy the product formula. In particular, prescribing $\gamma_v = j^2$ on a nonempty set $S \subseteq V_L$ forces $\gamma_v = j^2$ at every place. The constant global extension is admissible; extending the prescribed values by unequal complementary weights is impossible.*

Lemma 3.4 (Finite-fiber redistribution). *Let $\tau : V_L \rightarrow V_L$ have finite fibers (in particular, let τ be a permutation), let $a_v > 0$, and suppose the family $(b_{\tau(v)}^{a_v \gamma_v})_v$ satisfies the product formula for all $x \in L^\times$. Then the effective weights $\beta_u := \sum_{v: \tau(v)=u} a_v \gamma_v$ are constant in u .*

Proof. Reindexing the (for each fixed x finite) sum, $\sum_v a_v \gamma_v \log b_{\tau(v)}(x) = \sum_u \beta_u \log b_u(x) = 0$ for all $x \in L^\times$ — the hypothesis of Theorem 3.1 in the effective weights. Finiteness of the fibers makes each β_u a finite sum and the reindexing legitimate. $\square$

For infinite correspondences one would additionally need absolute summability per fiber and a justified rearrangement; no such generality is used in this paper, and none is needed for the audited passages, which move no places.

4 The Effective-Weight Quantity Class and the Exhaustion Proposition

Definition 4.1 (Effective-weight quantity class). Fix a domain Z . A family $Q = (Q_y : Z \rightarrow \mathbb{R})_y$ lies in $\mathcal{G}_{\text{eff}}$ if there are y -independent local functionals $q_w : Z \rightarrow \mathbb{R}$, $w \in V_L$, such that

$$Q_y(z) = \sum_{w \in V_L} \beta_{y,w} q_w(z), \quad Q_{\text{ref}}(z) := \sum_{w \in V_L} q_w(z).$$

Both sums must be finite or absolutely convergent, the first for every (y, z) and the reference sum for every z . The reference-sum requirement also covers parameters with $\beta_y = 0$.

Membership requires this *linear* representation; mere dependence on β_y is insufficient. For example, on a one-point domain take $\beta_{y,w} = c_y$ at every place, with c_y taking values 1 and 2, and $Q_y = c_y^2$. A representation would give $Q_y = c_y S$, where $S = \sum_w q_w$ is independent of y , forcing both $S = 1$ and $S = 2$. This counterexample to the earlier membership gloss is part of the R18 correction, not a new claim about a source height. The generic reference Q_{ref} need not be a classical arithmetic quantity; that name requires a separately identified height or degree and its domain.

Lemma 4.2 (Factorization). *Assume the fixed-field, full-place weighted product formula*

$$\prod_v \rho_{y,v}(x)^{\lambda_{y,v}} = 1 \quad \text{for every } x \in L^\times.$$

Then $\beta_{y,w} = c_y$ for all w , and every $Q \in \mathcal{G}_{\text{eff}}$ satisfies

$$Q_y = c_y Q_{\text{ref}}.$$

The scalar may depend on y . If (5.3.3) holds pointwise, then $\beta_y = 1$ and $Q_y = Q_{\text{ref}}$. Abstract real weights can have $c_y = 0$; for the source's positive $a_{y,v}$ and positive normalization exponents, $c_y > 0$.

Proof. For every $x \in L^\times$, the hypothesis becomes

$$0 = \sum_v \lambda_{y,v} \log \rho_{y,v}(x) = \sum_w \beta_{y,w} \log b_w(x).$$

Theorem 3.1 yields $\beta_{y,w} = c_y$. Substitution into Definition 4.1 gives the result; the specified convergence ensures the reference sum exists even when $c_y = 0$. $\square$

Proposition 4.3 (Exhaustion on the stated arithmetic domains). *For the projective height of ATS $II(\frac{1}{2})$, Definition 8.2.1 [5], on $\mathbb{P}^n(L)$, and the*

weighted degree on $L^\times$, use the fixed full standard family and the normalization identity (5.3.3). Then

$$\begin{aligned} h_y([x_0 : \cdots : x_n]) &= \sum_v \lambda_{y,v} \log \max_i \rho_{y,v}(x_i) \\ &= h_{\text{cl}}([x_0 : \cdots : x_n]), \\ \deg_y(x) &= \sum_v \lambda_{y,v} \log \rho_{y,v}(x) = \deg_{\text{cl}}(x). \end{aligned}$$

The degree equality has domain $x \in L^\times$.

Proof. On a nonzero representative tuple $(x_0, \dots, x_n) \in L^{n+1} \setminus \{0\}$, positivity of $a_{y,v}$ and $\beta_{y,v} = 1$ give

$$\begin{aligned} \sum_v \lambda_{y,v} \log \max_i \rho_{y,v}(x_i) &= \sum_v \lambda_{y,v} a_{y,v} \log \max_i b_v(x_i) \\ &= \sum_v \log \max_i b_v(x_i). \end{aligned}$$

For each tuple, these sums have finite support. Changing the representative by $u \in L^\times$ changes the standard sum by $\sum_v \log b_v(u) = 0$; the weighted sum has the same property by (5.3.4) and (5.3.3), as reflected in Lemma 8.2.3 of the source. Thus the equality descends to $\mathbb{P}^n(L)$. For the degree the calculation omits the maximum and is valid on $L^\times$. $\square$

The local height functionals $q_v(x_0, \dots, x_n) = \log \max_i b_v(x_i)$ are functions of representative tuples, not individually well-defined projective functions. They give a linear representation on the tuple domain. For a representation on $\mathbb{P}^n(L)$, one may first fix a y -independent representative for each point; the *summed* height is independent of that choice under the product formula. The degree uses $q_v(x) = \log b_v(x)$ on $L^\times$.

Remark 4.4 (Boundary and stabilization). This is a statement about these identified domains and the linear class, not every height or measure object in the ATS series. In ATS II($\frac{1}{2}$) §§8.6–8.7, the stabilized height is evaluated on $x \in L^\times$. Each summand $h_{\alpha,y}(x)$ equals the same classical value under the normalization identity, so its supremum equals that value. No additional exchange of supremum and sum is needed for this constant family. A general supremum of unclassified quantities has no such conclusion. Log-volumes, hulls, Haar normalizations, and Jacobians have neither membership nor nonmembership proved here; their concrete representation is the open measure-pinning obligation R-E2.

5 Source-Bound Application to the Height Sub-chain

Table 2 separates the direct substitution result from source objects requiring an additional bridge. Page anchors refer to the exact PDFs cited, not a claim

to audit their entire contents.

Table 2: Substitution results and unproved domain or object bridges.

Source site	Consequence within the stated contract	Limit
II($\frac{1}{2}$) (5.3.3)–(5.3.4), p. 32	$\beta_y = 1$; the weighted product formula reduces to the standard formula	Same L and full V_L
II($\frac{1}{2}$) Def. 8.2.1 / Lem. 8.2.3, p. 51	$h_y = h_{\text{cl}}$ on $\mathbb{P}^n(L)$	Representative tuples must descend
II($\frac{1}{2}$) Rem. 8.2.2 / Prop. 8.3.1, p. 51	The source extends to $\mathbb{P}^n(\overline{L})$ and asserts nontrivial dependence there	Extension, valuation and base-change bridge unproved here; not refuted by the L -domain identity alone
II($\frac{1}{2}$) §§8.6–8.7, pp. 54–55	On $L^\times$, every normalized summand equals h_{cl} , hence so does the supremum	No conclusion for other stabilized objects
III Rem. 8.9.2(2), p. 90	Comparison of heights of points in $X(\overline{L})$ using an ample line bundle is a separate source object	Bridge to the audited projective height remains open
IV §1.5, p. 17	The term-by-term restriction concerns lower and upper log-volume comparisons across y_0 and $\varphi(y_0)$	Not a universal prohibition of all local comparisons; no $\mathcal{G}_{\text{eff}}$ classification
II($\frac{1}{2}$) Thm. 5.10.1(7)	Coordinate kernels can differ for nonproportional coefficients while their $L^\times$ pullbacks coincide	Conditional interpretation, not verification of the whole source theorem

Applicability gate H-PLACES. The theorem uses one fixed number field, all its archimedean and finite places, and all nonzero elements of that field as tests. A source notation that ranges over nonarchimedean places (such as ATS III §3.2(5)) cannot automatically be identified with this V_L . Likewise, places of an extension L' tested only on $L^\times$ require an explicit field-descent and place bridge. Neither indexing identification is supplied by a coordinate analogy.

The source itself supplies the pointwise normalization used in the L -domain substitution. This establishes the displayed height and degree identities, but does not identify every operative height, line-bundle quantity, or log-volume elsewhere in the series.

6 Conditional Consequences (Three Typed Cases)

The cases concern a verified quantity representation and an explicitly matched domain, field, place family, and test set.

Case A (pointwise identity (5.3.3)).

Then $\beta_y = 1$ and every proved member of $\mathcal{G}_{\text{eff}}$ equals its reference. The identified height on $\mathbb{P}^n(L)$ and degree on $L^\times$ are classical and y -independent. Application to another operative quantity requires identifying that object and its domain bridge.

Case B (full weighted product formula only).

Then $\beta_y = c_y \mathbf{1}$ and $Q_y = c_y Q_{\text{ref}}$ for $Q \in \mathcal{G}_{\text{eff}}$. The scalar can vary with y , but final weights cannot vary from place to place. For actual standard heights and degrees, this is a global rescaling on their respective domains.

Case C (membership unclassified).

For the concrete log-volumes, hulls, and measures no linear representation or obstruction to one is proved here. The theorem still constrains effective weights whenever its weighted product-formula hypothesis holds; it does not thereby classify these quantity families. R-E2 remains open.

Table 3: Typed consequence matrix.

Case	Effective weights	Global quantities	Derivation burden
A: pointwise identity	$\beta_y = 1$	$Q_y = Q_{\text{ref}}$ for proved members	Identify operative object and domain bridge
B: full weighted formula	$\beta_y = c_y \mathbf{1}$	$Q_y = c_y Q_{\text{ref}}$ for proved members	Check weighted-formula hypothesis, indexing and factorization if another variation is intended
C: unclassified quantity	Constrained if the same theorem hypotheses hold	No classification of the quantity from weights alone	Settle the concrete representation and measure-pinning bridge

Conditional consequence. A place-selective metric variation cannot be carried by the verified linear L -domain families under the full weighted formula; the pointwise identity removes even their global rescaling. Prop. 8.3.1 on $\mathbb{P}^n(\overline{L})$, the ATS III line-bundle comparison, and ATS IV's log-volume apparatus are separate targets. Their operative object and transport bridges

must be checked before applying this conclusion. The standard product formula for the fixed b is not something these cases call on a source to disprove. No global collapse, closure of the repair program, or exclusion of other routes follows.

Falsification and revision criteria.

1. A nonconstant effective-weight family satisfying exactly the same-field, full-place product formula for every $x \in L^\times$ would contradict Theorem 3.1. A counterexample to factorization must also satisfy the linear representation and convergence contract.
2. Evidence that an operative source height has another domain or definition defeats an unproved identification step, not the theorem. This is why the extension and line-bundle bridges remain separate.
3. A future source revision may supply new objects or bridges. It would require a new audit of that version; it neither retroactively proves an absent bridge nor refutes a statement tied to the cited version.

7 External Program Status (Not Part of the Present Proofs)

Part 5 is published [15]; its publication does not discharge the remaining mathematical bridges. R6-1 / REP-6-1 is a conditional algebraic box-transport reconstruction: actual Haar normalization, tensor/index compatibility and the operative volume comparison remain separate tasks. A defect in an intermediate weighted model is not by itself a counterexample to a Joshi theta-volume theorem. R6-2 / REP-6-2 assumes V1 and an explicit $L' \rightarrow L$ place bridge. R6-3 / REP-6-3 concerns uniqueness of the same-field normalized data, not equality of all global coordinate hyperplanes. These notices refer to the August program and its R18 corrections; none is used in the proofs above.

The GNC source audit is cited in version 1.2 [12], and the Part 4 ledger audit in version 0.3 [11]. The certificate-form criterion is a separate instrument [14]. Part 6's R-E2 measure-pinning obligation is not GNC-E2. Likewise REP-6-1–3 are repair positions, not ledger F-R identifiers; publication version 0.3 is not a claim to a new v1-F ledger acceptance. The P5/Shadow Tomograph and full-chain obligations remain open.

8 Claim Boundaries

Theorem 3.1 and Lemma 3.4 concern the explicit fixed-field, full-place contract. Lemma 4.2 additionally requires a proved linear representation; Proposition 4.3 concerns $\mathbb{P}^n(L)$ and $L^\times$. Only these verified quantity/domain pairs

support the negative statement about residual place-selective variation. The $\overline{L}$ extension, ATS III line-bundle heights and ATS IV log-volumes have not been identified with that contract here. No exhaustive absence of another metric carrier, proof of Case C nonmembership, closure of the repair program, or whole-source error verdict is asserted. The coordinate-kernel interpretation is conditional and does not certify the full nonconstancy assertion of source Theorem 5.10.1(7). There is no proof or refutation of abc or IUT and no whole-series mathematical acceptance.

The R18 inventory is the source of these corrections. Independent manuscript and source checks are review evidence, not a machine-checked Lean proof, a numerical experiment, or a new formal verification result.

AI disclosure

AI systems assisted in drafting, translation, and review of this working paper. The corrective version uses the R18 inventory and independently checked manuscript and selected source passages. The author remains responsible for the text and its claims. These checks do not constitute formal proof verification or mathematical acceptance of the whole ATS series. No machine-checked proof or new numerical experiment is reported.

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